Exponential Functions

Exponential Functions

The function $f(x) = 2^x$ is called an *exponential function* because the variable, x, is the exponent. It should not be confused with the power function $g(x) = x^2$, in which the variable is the base.

In general, an **exponential function** is a function of the form

$$f(x) = b^x$$

where b is a positive constant. Let's recall what this means. If $x = n$, a positive integer, then

$$b^n = \underbrace{b \cdot b \cdot \cdots \cdot b}_{n \text{ factors}}$$

Exponential Functions

If $x = 0$, then $b^0 = 1$, and if $x = -n$, where n is a positive integer, then

$$b^{-n} = \frac{1}{b^n}$$

If x is a rational number, $x = p/q$, where p and q are integers and $q > 0$, then

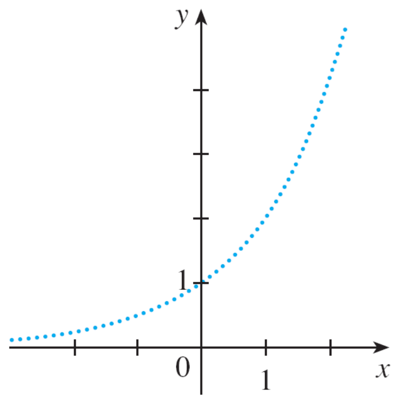
$$b^x = b^{p/q} = \sqrt[q]{b^p} = \left(\sqrt[q]{b}\right)^p$$

But what is the meaning of b^x if x is an irrational number? For instance, what is meant by $2^{\sqrt{3}}$ or 5^{π}?

Exponential Functions

To help us answer this question we first look at the graph of the function $y = 2^x$, where x is rational. A representation of this graph is shown in Figure 1.

We want to enlarge the domain of $y = 2^x$ to include both rational and irrational numbers.



Representation of $y = 2^x$, x rational

Figure 1

There are holes in the graph in Figure 1 corresponding to irrational values of x.

We want to fill in the holes by defining $f(x) = 2^x$, where $x \in \mathbb{R}$, so that f is an increasing function.

Exponential Functions

In particular, since the irrational number $\sqrt{3}$ satisfies

$$1.7 < \sqrt{3} < 1.8$$

we must have

$$2^{1.7} < 2^{\sqrt{3}} < 2^{1.8}$$

and we know what $2^{1.7}$ and $2^{1.8}$ mean because 1.7 and 1.8 are rational numbers.

Exponential Functions

Similarly, if we use better approximations for $\sqrt{3}$, we obtain better approximations for $2^{\sqrt{3}}$.

$$
\begin{array}{rcl}
1.73 < \sqrt{3} < 1.74 & \Rightarrow & 2^{1.73} < 2^{\sqrt{3}} < 2^{1.74} \\
1.732 < \sqrt{3} < 1.733 & \Rightarrow & 2^{1.732} < 2^{\sqrt{3}} < 2^{1.733} \\
1.7320 < \sqrt{3} < 1.7321 & \Rightarrow & 2^{1.7320} < 2^{\sqrt{3}} < 2^{1.7321} \\
1.73205 < \sqrt{3} < 1.73206 & \Rightarrow & 2^{1.73205} < 2^{\sqrt{3}} < 2^{1.73206} \\
\vdots \quad \vdots & & \vdots \quad \vdots
\end{array}
$$

Exponential Functions

It can be shown that there is exactly one number that is greater than all of the numbers

$$2^{1.7},\quad 2^{1.73},\quad 2^{1.732},\quad 2^{1.7320},\quad 2^{1.73205},\quad \ldots$$

and less than all of the numbers

$$2^{1.8},\quad 2^{1.74},\quad 2^{1.733},\quad 2^{1.7321},\quad 2^{1.73206},\quad \ldots$$

We define $2^{\sqrt{3}}$ to be this number. Using the preceding approximation process we can compute it correct to six decimal places:

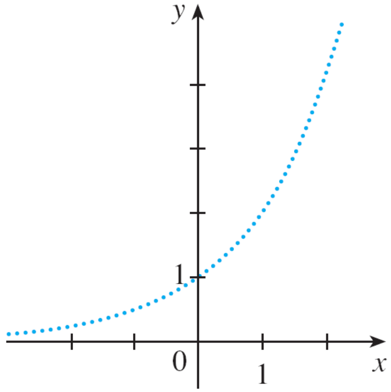
$$2^{\sqrt{3}} \approx 3.321997$$

Exponential Functions

Similarly, we can define 2^x (or b^x, if $b > 0$) where x is any irrational number.

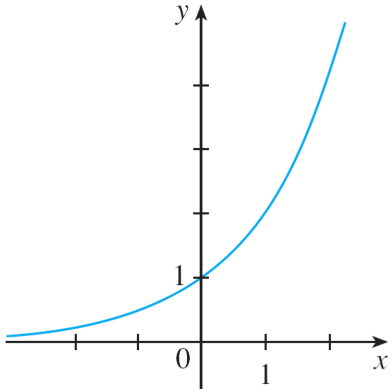
Figure 2 shows how all the holes in Figure 1 have been filled to complete the graph of the function

$$f(x) = 2^x,\ x \in \mathbb{R}.$$



Representation of $y = 2^x$, x rational

Figure 1



$y = 2^x$, x real

Figure 2

Exponential Functions

The graphs of members of the family of functions $y = b^x$ are shown in Figure 3 for various values of the base b.

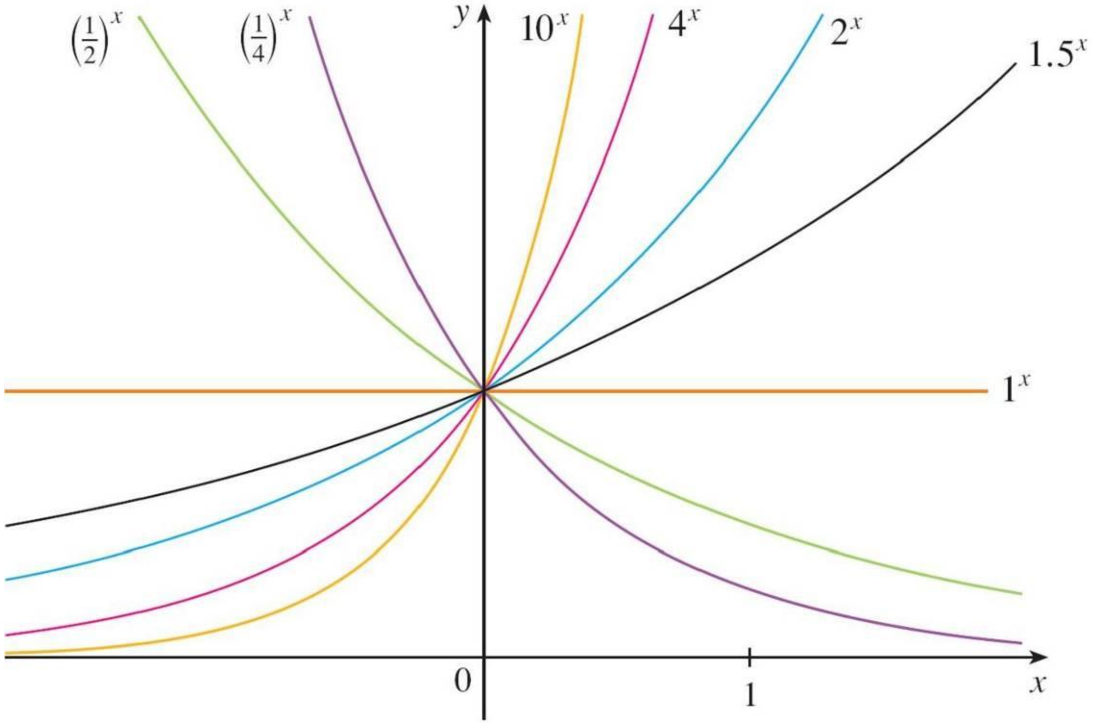


Figure 3

Exponential Functions

Notice that all of these graphs pass through the same point (0, 1) because $b^0 = 1$ for $b \neq 0$.

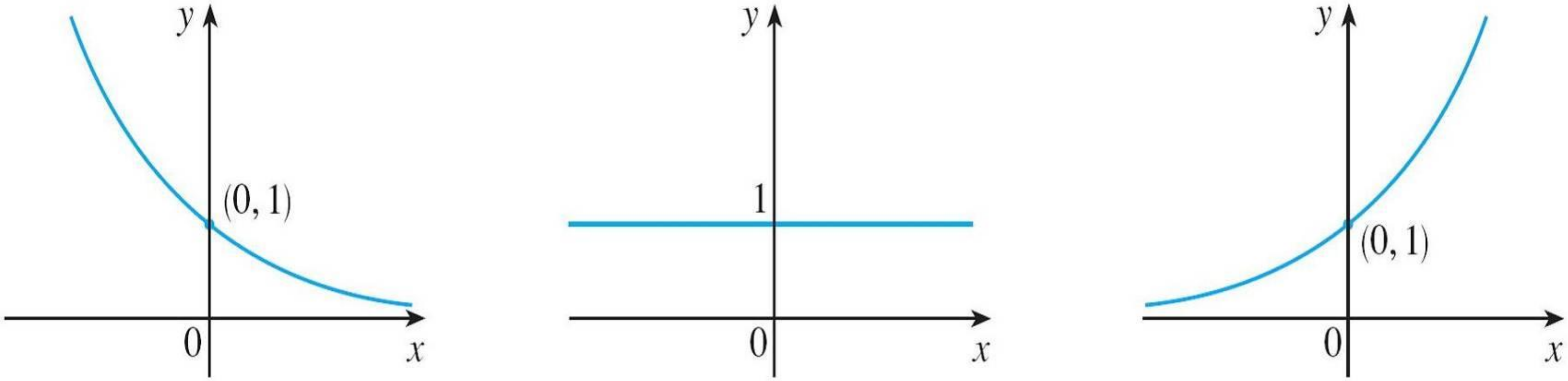
Notice also that as the base b gets larger, the exponential function grows more rapidly (for $x > 0$).

You can see from Figure 3 that there are basically three kinds of exponential functions $y = b^x$.

If $0 < b < 1$, the exponential function decreases; if $b = 1$, it is a constant; and if $b > 1$, it increases.

Exponential Functions

These three cases are illustrated in Figure 4.



(a) $y = b^x$, $0 < b < 1$ **(b)** $y = 1^x$ **(c)** $y = b^x$, $b > 1$

Figure 4

Exponential Functions

Observe that if $b \neq 1$, then the exponential function $y = b^x$ has domain $\mathbb{R}$ and range $(0, \infty)$.

Notice also that, since $(1/b)^x = 1/b^x = b^{-x}$, the graph of $y = (1/b)^x$ is just the reflection of the graph of $y = b^x$ about the y-axis.

Exponential Functions

One reason for the importance of the exponential function lies in the following properties.

If x and y are rational numbers, then these laws are well known from elementary algebra. It can be proved that they remain true for arbitrary real numbers x and y.

Laws of Exponents If a and b are positive numbers and x and y are any real numbers, then

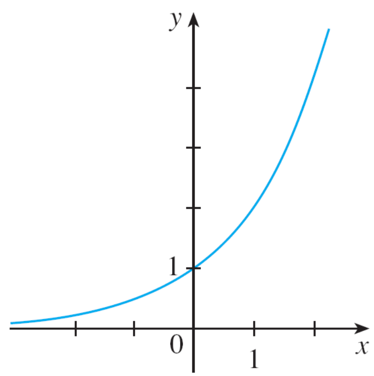
1. $b^{x+y} = b^x b^y$ **2.** $b^{x-y} = \dfrac{b^x}{b^y}$ **3.** $(b^x)^y = b^{xy}$ **4.** $(ab)^x = a^x b^x$

Example 1

Sketch the graph of the function $y = 3 - 2^x$ and determine its domain and range.

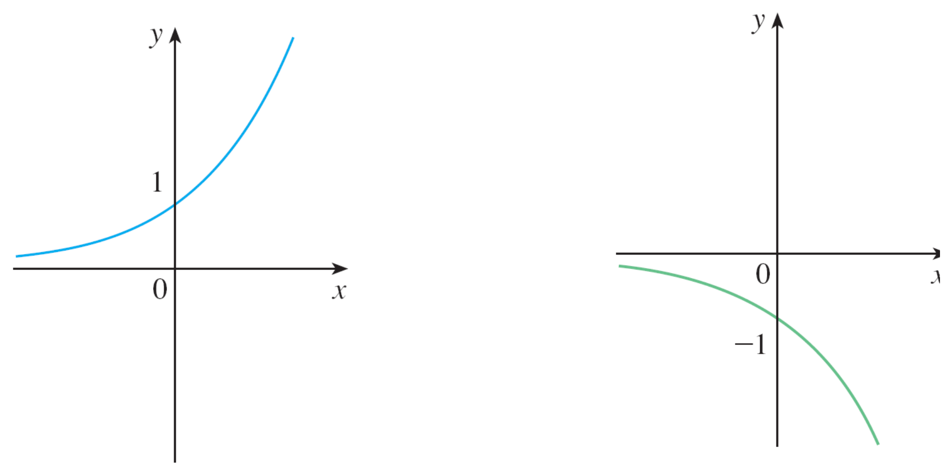
Solution:

First we reflect the graph of $y = 2^x$ [shown in Figures 2 and 5(a)] about the x-axis to get the graph of $y = -2^x$ in Figure 5(b).



$y = 2^x$, x real

Figure 2



(a) $y = 2^x$

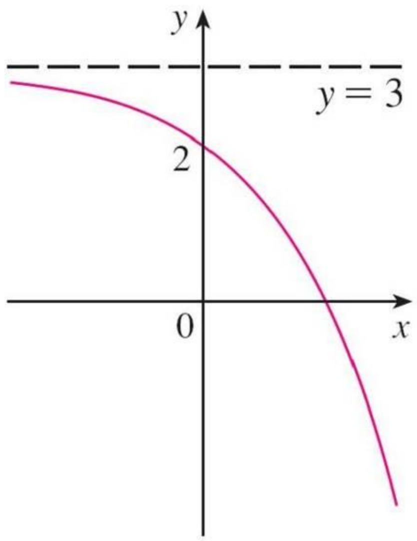
(b) $y = -2^x$

Figure 5

Example 1 – *Solution*

cont'd

Then we shift the graph of $y = -2^x$ upward 3 units to obtain the graph of $y = 3 - 2^x$ in Figure 5(c).



(c) $y = 3 - 2^x$

Figure 5

The domain is $\mathbb{R}$ and the range is $(-\infty, 3)$.

Applications of Exponential Functions

Applications of Exponential Functions

The exponential function occurs very frequently in mathematical models of nature and society. Here we indicate briefly how it arises in the description of population growth and radioactive decay.

First we consider a population of bacteria in a homogeneous nutrient medium. Suppose that by sampling the population at certain intervals it is determined that the population doubles every hour.

Applications of Exponential Functions

If the number of bacteria at time t is $p(t)$, where t is measured in hours, and the initial population is $p(0) = 1000$, then we have

$$p(1) = 2p(0) = 2 \times 1000$$

$$p(2) = 2p(1) = 2^2 \times 1000$$

$$p(3) = 2p(2) = 2^3 \times 1000$$

It seems from this pattern that, in general,

$$p(t) = 2^t \times 1000 = (1000)2^t$$

Applications of Exponential Functions

This population function is a constant multiple of the exponential function $y = 2^t$, so it exhibits the rapid growth.

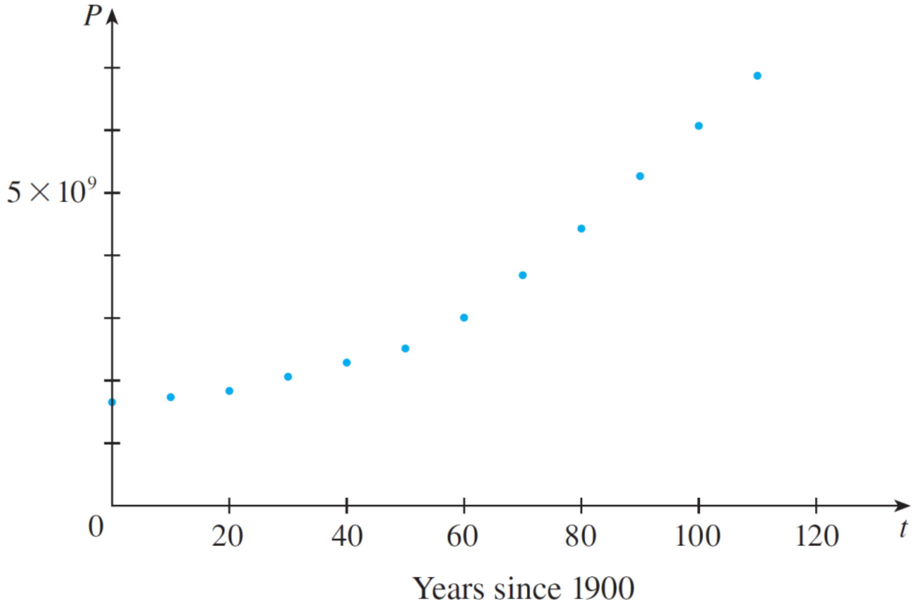
Under ideal conditions (unlimited space and nutrition and absence of disease) this exponential growth is typical of what actually occurs in nature.

Applications of Exponential Functions

What about the human population? Table 1 shows data for the population of the world in the 20th century and Figure 8 shows the corresponding scatter plot.

Table 1

t (years since 1900)	Population (millions)
0	1650
10	1750
20	1860
30	2070
40	2300
50	2560
60	3040
70	3710
80	4450
90	5280
100	6080
110	6870



Scatter plot for world population growth

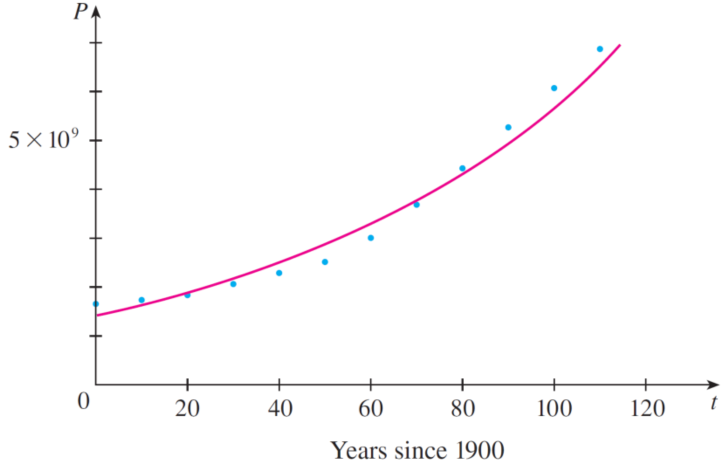
Figure 8

Applications of Exponential Functions

The pattern of the data points in Figure 8 suggests exponential growth, so we use a graphing calculator with exponential regression capability to apply the method of least squares and obtain the exponential model

$$P = (1436.53) \cdot (1.01395)^t$$

where $t = 0$ corresponds to 1900. Figure 9 shows the graph of this exponential function together with the original data points.



Exponential model for population growth

Figure 9

Applications of Exponential Functions

We see that the exponential curve fits the data reasonably well.

The period of relatively slow population growth is explained by the two world wars and the Great Depression of the 1930s.

Example 3

The half-life of strontium-90, ^{90}Sr, is 25 years. This means that half of any given quantity of ^{90}Sr will disintegrate in 25 years.

(a) If a sample of ^{90}Sr has a mass of 24 mg, find an expression for the mass $m(t)$ that remains after t years.

(b) Find the mass remaining after 40 years, correct to the nearest milligram.

(c) Use a graphing device to graph $m(t)$ and use the graph to estimate the time required for the mass to be reduced to 5 mg.

Example 3 – *Solution*

(a) The mass is initially 24 mg and is halved during each 25-year period, so

$$m(0) = 24$$

$$m(25) = \frac{1}{2}(24)$$

$$m(50) = \frac{1}{2} \cdot \frac{1}{2}(24) = \frac{1}{2^2}(24)$$

$$m(75) = \frac{1}{2} \cdot \frac{1}{2^2}(24) = \frac{1}{2^3}(24)$$

$$m(100) = \frac{1}{2} \cdot \frac{1}{2^3}(24) = \frac{1}{2^4}(24)$$

Example 3 – *Solution*

cont'd

From this pattern, it appears that the mass remaining after *t* years is

$$m(t) = \frac{1}{2^{t/25}}(24) = 24 \cdot 2^{-t/25} = 24 \cdot (2^{-1/25})^t$$

This is an exponential function with base

$$b = 2^{-1/25} = 1/2^{1/25}.$$

(b) The mass that remains after 40 years is

$$m(40) = 24 \cdot 2^{-40/25} \approx 7.9 \text{ mg}$$

Example 3 – *Solution*

cont'd

(c) We use a graphing calculator or computer to graph the function $m(t) = 24 \cdot 2^{-t/25}$ in Figure 12.

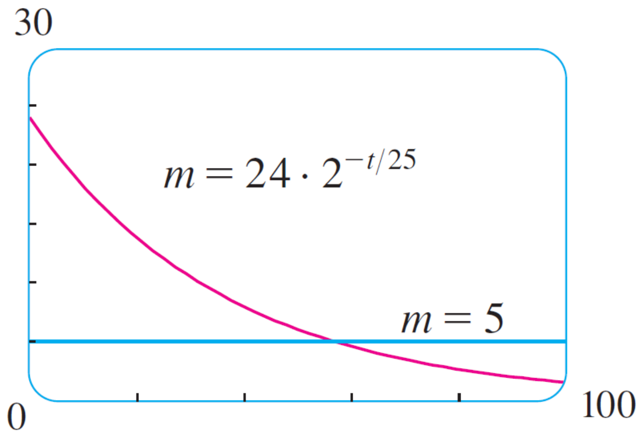


Figure 12

We also graph the line $m = 5$ and use the cursor to estimate that $m(t) = 5$ when $t \approx 57$. So the mass of the sample will be reduced to 5 mg after about 57 years.

The Number *e*

The Number e

Of all possible bases for an exponential function, there is one that is most convenient for the purposes of calculus.

The choice of a base b is influenced by the way the graph of $y = b^x$ crosses the y-axis.

The Number e

Figures 13 and 14 show the tangent lines to the graphs of $y = 2^x$ and $y = 3^x$ at the point $(0, 1)$.

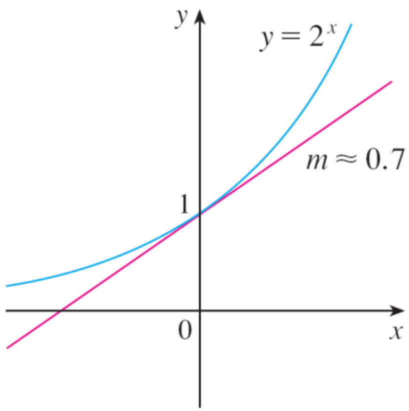


Figure 13

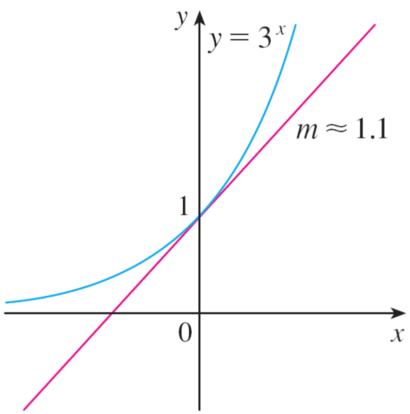
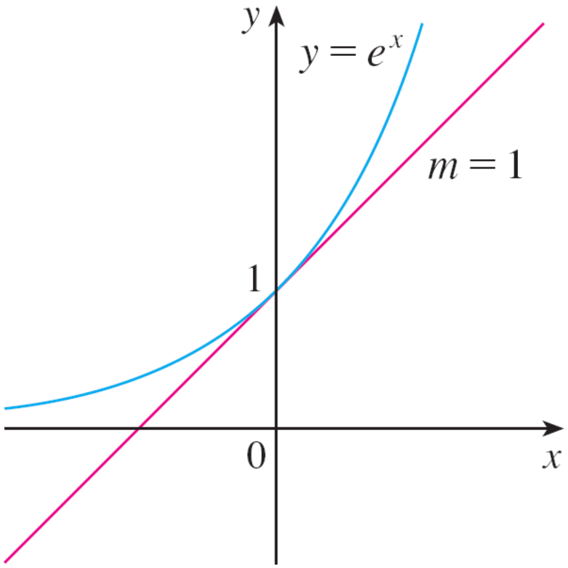


Figure 14

(For present purposes, you can think of the tangent line to an exponential graph at a point as the line that touches the graph only at that point.) If we measure the slopes of these tangent lines at (0, 1), we find that $m \approx 0.7$ for $y = 2^x$ and $m \approx 1.1$ for $y = 3^x$.

The Number e

It turns out, that some of the formulas of calculus will be greatly simplified if we choose the base b so that the slope of the tangent line to $y = b^x$ at (0, 1) is *exactly* 1.
(See Figure 15.)



The natural exponential function crosses the *y*-axis with a slope of 1.

Figure 15

The Number *e*

In fact, there *is* such a number and it is denoted by the letter *e*. (This notation was chosen by the Swiss mathematician Leonhard Euler in 1727, probably because it is the first letter of the word *exponential*.)

The Number e

In view of Figures 13 and 14, it comes as no surprise that the number e lies between 2 and 3 and the graph of $y = e^x$ lies between the graphs of $y = 2^x$ and $y = 3^x$.
(See Figure 16.)

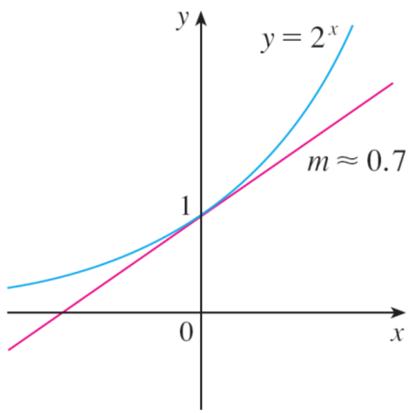


Figure 13

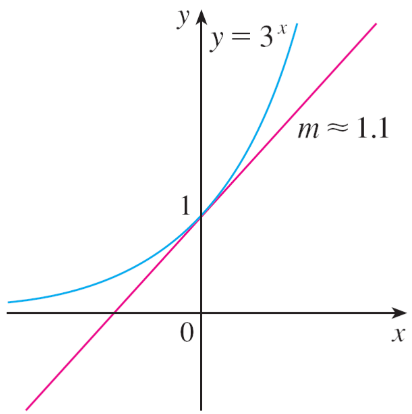


Figure 14

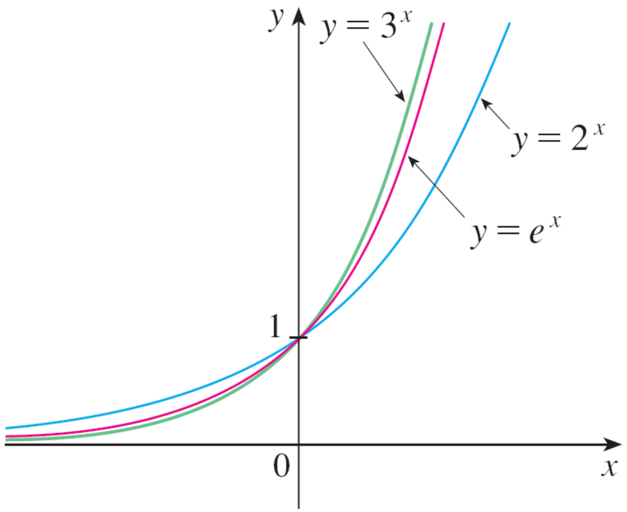


Figure 16

The Number e

We will see that the value of e, correct to five decimal places, is

$$e \approx 2.71828$$

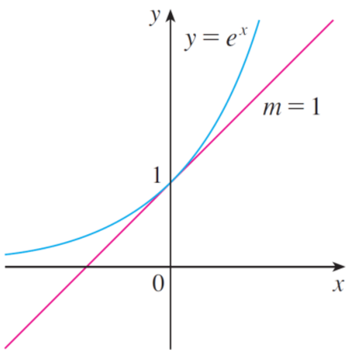
We call the function $f(x) = e^x$ the **natural exponential function**.

Example 4

Graph the function $y = \frac{1}{2}e^{-x} - 1$ and state the domain and range.

Solution:

We start with the graph of $y = e^x$ from Figures 15 and 17(a) and reflect about the *y*-axis to get the graph of $y = e^{-x}$ in Figure 17(b). (Notice that the graph crosses the *y*-axis with a slope of –1).



The natural exponential function crosses the *y*-axis with a slope of 1.

Figure 15

(a) $y = e^x$

(b) $y = e^{-x}$

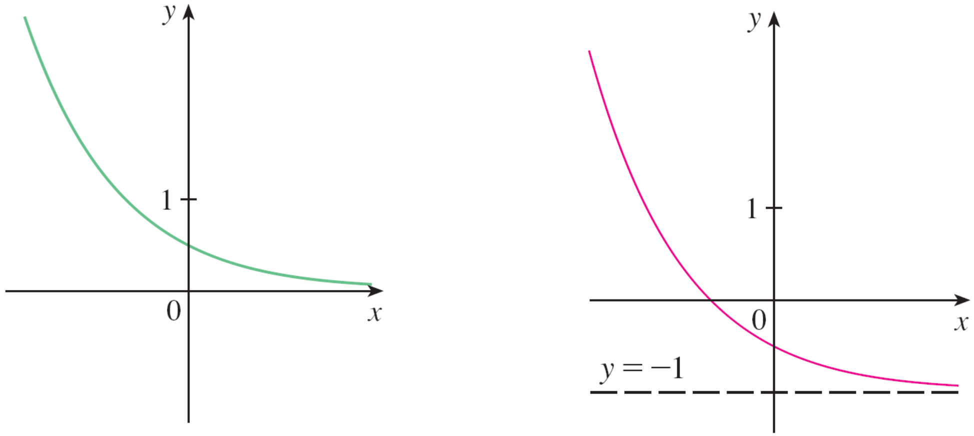
Figure 17

Example 4 – *Solution*

cont'd

Then we compress the graph vertically by a factor of 2 to obtain the graph of $y = \frac{1}{2}e^{-x}$ in Figure 17(c).

Finally, we shift the graph downward one unit to get the desired graph in Figure 17(d).



(c) $y = \frac{1}{2}e^{-x}$

(d) $y = \frac{1}{2}e^{-x} - 1$

Figure 17

The domain is $\mathbb{R}$ and the range is $(-1, \infty)$.